Whole weighted-Proj integrality under a separable quintic

Formal proof packet

28 October 2026

Pinned Lean 4.34.1 checks that the actual weighted-Proj curve is integral for any field and coefficient tuple when its quintic is separable. The direct axiom audit of the three new theorems lists only `propext`, `Classical.choice`, and `Quot.sound`. The theorem has no additional characteristic restriction.

Checked proof structure

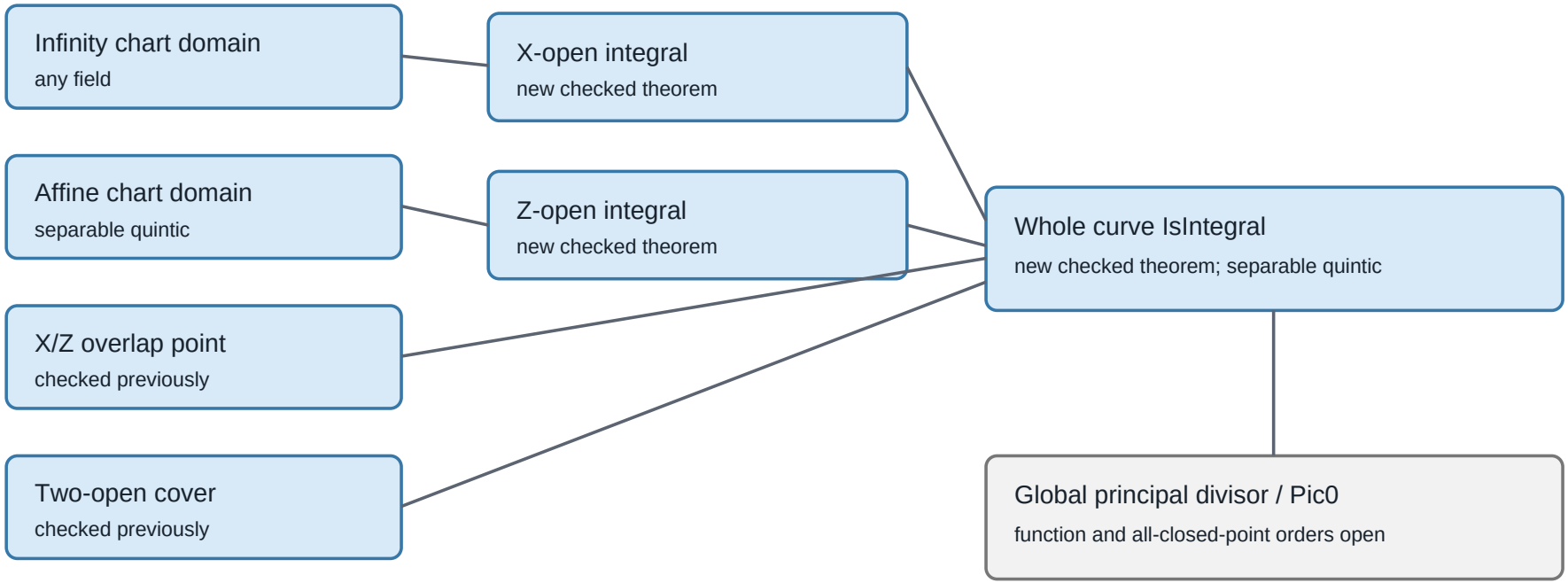
The infinity-chart ring is a domain for every field, and the affine-chart ring is a domain under separability. Two new theorems transport these facts through the checked scheme isomorphisms to the actual X- and Z-basic opens. The previously checked overlap point and two-open cover supply a pairwise nonempty open cover. The whole-curve theorem applies pinned Mathlib's gluing of preirreducibility and reducedness, then its equivalence of integral schemes with reduced irreducible schemes.

Scope boundary

This establishes the scheme-level integrality prerequisite for the function-field API. The source-style common 2+2 function has not been constructed in that field. Its orders over all closed points, principal divisor, Picard-class equality, and compiled Rust correspondence remain open. The accompanying CSV gives each exact status; the Lean source and compiler log give the theorem statements and axiom output. No DLP experiment or cost comparison was run.

Proof dependencies

Whole weighted-Proj integrality proof



Blue = checked declarations. Gray = open construction. Arrows = proof dependencies, not measurements.

Blue nodes are checked declarations; gray is open construction. Arrows represent proof dependencies, not performance measurements.